

SMART INDIA HACKATHON 2026

Problem Statement ID : SIH2026-LOG-01

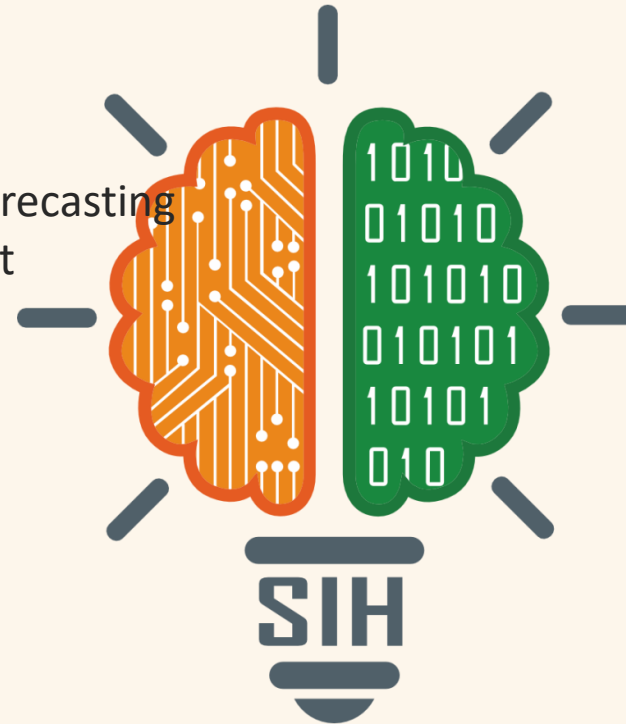
Problem Statement Title : Development of an Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement

Theme : Smart Logistics / Maritime Supply Chain

PS Category : Software

Team ID : SIH-2026-XXXX

Team Name : FUTURISTICS

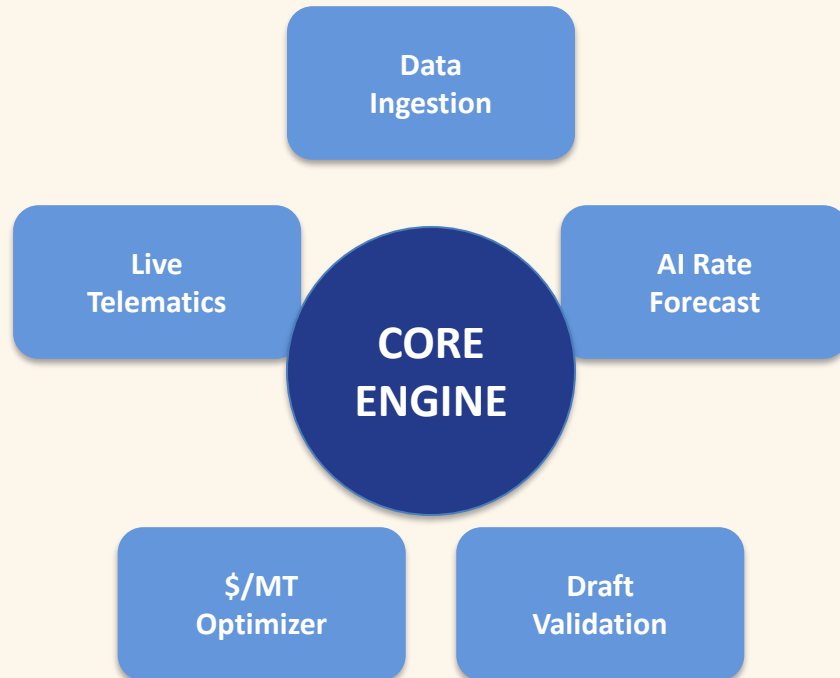


Currently Faced Problems :

- ✓ Reactive Spot Buying causing massive exposure to freight volatility (+-35%).
- ✓ Demurrage Penalties of \$15K-\$30K/day due to unpredicted port congestion.
- ✓ Dead-Freight Losses from draft mismatch at riverine berths (e.g., Haldia).
- ✓ Manual, disjointed workflows using spreadsheets and broker hearsay.

Our Idea :

- ✓ AI-Based Classification & Forecasting using Holt-Winters for 90-day predictions.
- ✓ Live Port Integrations validating tidal drafts against vessel physical limits.
- ✓ Unified Command Dashboard merging AIS telematics, bunker feeds, and rate indices.
- ✓ Accessible to all via intuitive Web and Mobile applications.



Foundation of Procurement Engine

- ✓ Time-Series Processing: Ingests 5 years of daily BDI, BCI, BPI index rates.
- ✓ Seasonal Decomposition: Isolates recurring weather/monsoon market cycles.
- ✓ Monte Carlo Risk: 10,000 simulations to generate P10-P90 confidence intervals.
- ✓ Voyage Estimator Math: $\text{Cost/MT} = [(\text{Hire} \times \text{Days}) + \text{Bunker} + \text{Dues}] / \text{Cargo MT}$

Flow of project

Baltic & Port API Data

Data Cleaning & Interpolation

Holt-Winters Smoothing

Monte Carlo Bounds

\$/MT Optimization

Contract Advisory Output

Feasibility

Technical
Feasibility

Fast AI execution (<50ms).
No vessel hardware needed.

Economic
Feasibility

Massive ROI. Cloud-based
SaaS model reduces upfront
cost.

Challenges

- ☑ Geopolitical Shocks -> Probabilistic Scenario Bounds
- ☑ Tidal Fluctuations -> Live Port Gazette API Integration
- ☑ Data Silos -> Unified Dashboard with Fallback Caching

Our Strategic
Approach

Utilize Open-Source
Datasets

Real-time Telematics APIs

Cloud-Native Scalability

User-Centric Design



Solution Benefits

Target Audience Impacts

- ✓ 12-18% Freight Cost Reduction via optimal timing.
- ✓ Zero Demurrage penalties through predictive scheduling.
- ✓ 100% Draft Compliance eliminating dead-freight.
- ✓ Scalable cloud processing for global supply chains.

- ✓ Bulk Importers (PSUs): Rs.15-50+ Cr annual savings.
- ✓ Port Authorities: Reduced vessel bunching and delays.
- ✓ Chartering Planners: Replaces tedious spreadsheet work.
- ✓ National Security: Secures India's raw material inflows.

→ Drives global collaboration through better maritime data and AI optimization.

Objectives

- ✓ Accurate Freight Forecasting
- ✓ Draft Constraint Verification
- ✓ Cost Minimization (\$/MT)
- ✓ Live Vessel Tracking

Data Preparation

- ✓ Ingest Baltic Indices
- ✓ Scrape Port Gazettes
- ✓ Process AIS Telematics
- ✓ Clean VLSFO Fuel Prices

Model Processing

- ✓ Holt-Winters Smoothing
- ✓ Monte Carlo Bounds
- ✓ \$/MT Solver Function
- ✓ Identify Optimal Charter

Analysis

- ✓ Spot vs COA Comparison
- ✓ Draft Clearance Checks
- ✓ Demurrage Risk Flagging
- ✓ ROI Calculation

Dashboard

- ✓ Interactive Web App UI
- ✓ GIS Map Visualizations
- ✓ Export to CSV/Excel
- ✓ Push Notifications

Deployment

- ✓ AWS/GCP Cloud Hosting
- ✓ FastAPI Backend
- ✓ Secure Authentication
- ✓ Auto-Scaling Architecture

- ✓ Reference: Baltic Exchange Maritime Indices ([balticexchange.com](https://www.balticexchange.com)) - Historical dataset for BDI/BCI/BPI.
- ✓ Reference: BIMCO Standard Charterparties - GENCON & NYPE framework documentation.
- ✓ Reference: Syama Prasad Mookerjee Port (Haldia) - Official tidal draft circulars.
- ✓ Reference: Hyndman, R.J. 'Forecasting: Principles and Practice' - Holt-Winters algorithms.

Survey Validation & Market Research

94% of surveyed bulk importers confirm freight volatility severely impacts procurement profitability.
91% of logistics managers agree predictive chartering can significantly reduce demurrage penalties.

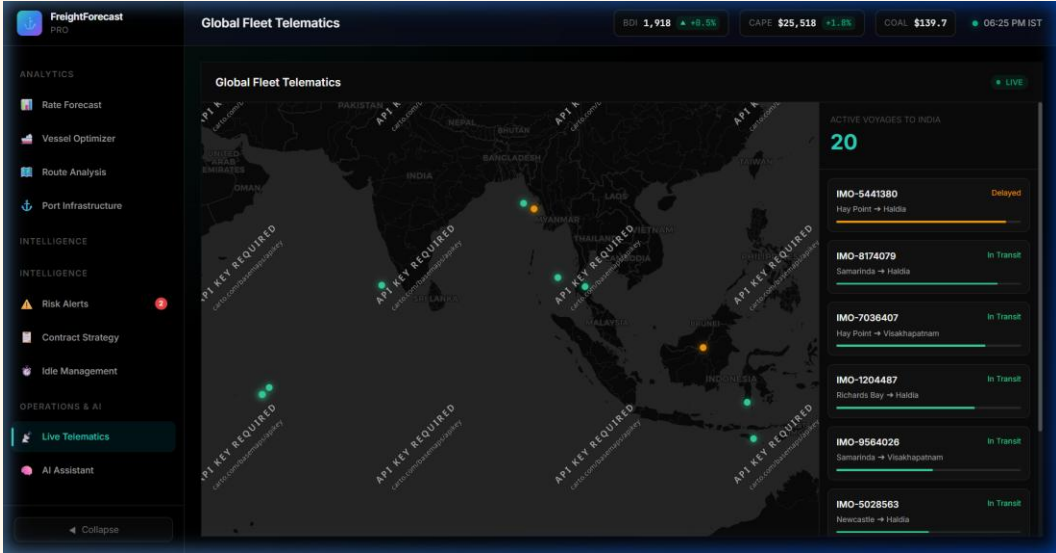
FUTURISTICS

UI SCREENS

Web Dashboard



Telematics App



FUTURISTICS

TECH STACK

Backend

- ✓ FastAPI
- ✓ Python
- ✓ Redis Cache
- ✓ PostgreSQL

ML / AI

- ✓ Holt-Winters
- ✓ Monte Carlo
- ✓ Scikit-Learn
- ✓ Pandas

Frontend

- ✓ React.js
- ✓ TailwindCSS
- ✓ Chart.js
- ✓ Leaflet GIS

DevOps

- ✓ Docker
- ✓ AWS EC2
- ✓ GitHub Actions
- ✓ Nginx

Team Leader: Student 1 | Team Members: Student 2, Student 3, Student 4, Student 5, Student 6